

IKEA Retail Sales Analysis

A SQL + Python Case Study on Retail Transaction Data

1. Project Overview

This project analyses a retail transaction dataset using MySQL and Python to uncover customer behaviour, product performance, sales trends, and revenue-loss drivers. The objective was not just to write queries, but to answer the questions a retail business would actually care about:

- When do customers shop, and does that pattern suggest anything about who the customers actually are?
- Who are the most valuable customers, and how should they be segmented?
- Which products drive revenue versus volume, and which are quietly costing the business through returns?
- How much revenue is lost to cancellations, and where is that risk concentrated?
- How does performance vary by geography, and where is growth opportunity concentrated?

2. Dataset & Data Cleaning

The raw dataset contained invoice-level transaction lines: invoice number, date, time, stock code, product description, quantity (negative values indicating returns/cancellations), unit price, customer ID, and country. Before analysis, the following cleaning steps were required:

- Parsed the combined invoice date-time field into separate date and time columns and converted to MySQL-compatible formats.
- Handled NULL and blank values encountered during CSV import.
- Kept invoice number as VARCHAR rather than INT, since cancelled invoices are prefixed with 'C' and would otherwise be lost.
- Treated negative quantity values as returns/cancellations throughout, rather than excluding them, this is what made the returns analysis possible.
- Formatting artifact: Country values carry a trailing carriage-return character (e.g. "United Kingdom\r") from the source file's line endings, cosmetic, but should be trimmed before any country field is used in a customer-facing report.
- Non-product entries: A handful of "description" values are not real products, Manual, POSTAGE, DOTCOM POSTAGE, Bank Charges, Discount, SAMPLES, Damaged, and check are internal adjustment or admin entries. Left in, they distort "top product" rankings (e.g. Manual appears as both the single highest-priced line item and among the most frequently returned). These were identified and should be excluded from any product-level ranking.
- Unidentified customers: custid = 0 represents guest or unrecorded-identity transactions. These were excluded from customer-level analysis (segmentation, CLV, VIP ranking), but the fact that they account for a large pooled revenue figure on their own is itself a data-collection gap worth flagging to the business.

3. Methodology

65 SQL queries were written across seven analytical themes, sales performance, product performance, customer behaviour, geography, returns/cancellations, order patterns and advanced analytics (RFM, segmentation,

cohorts) followed by exploratory data analysis in Python/Pandas for validation and visualization. Results were exported programmatically from MySQL into a single structured workbook to support consistent, repeatable analysis rather than one-off querying.

4. Key Business Metrics at a Glance

Metric	Value
Total revenue (valid transactions)	£10,644,560
Average order value	£513.54
Unique customers	4,373
Average spend per customer	£2,229
Orders cancelled	19.97% of invoices
Revenue lost to cancellations	£896,812 (~8.4% of gross revenue)
Repeat purchase rate	69.97%
One-time buyers	1,313 customers (30.0%)
Top 5% customers' share of revenue	37.8% (218 customers, £4.03M)
Top revenue country	United Kingdom (76.9% of revenue)

5. Key Business Insights

5.1 Sales Performance & Timing

- **Seasonality:** Revenue is heavily concentrated in November and December (£1.51M and £1.46M respectively), a pre-holiday peak consistent with gift and homeware demand.
- **Weekday concentration:** Weekday revenue (£9.83M) dwarfs weekend revenue (£813K) — a >12:1 ratio. Thursday and Tuesday are the strongest days.
- **Business-hours pattern:** Sales cluster almost entirely between 8am and 5pm, peaking at 10am, 12pm, and 3pm, with essentially no evening or night activity.

Taken together, the near-total absence of weekend and evening activity, combined with very high per-order quantities, is a strong behavioural signal that a meaningful share of this customer base is buying on a business/wholesale cadence rather than as individual end consumers, which shape how the segmentation and marketing recommendations are read.

5.2 Product Performance

- **By volume,** PAPER CRAFT LITTLE BIRDIE (80,995 units) and MEDIUM CERAMIC TOP STORAGE JAR (78,033 units) lead. By revenue — after excluding the non-product entries flagged in Section 2 — REGENCY CAKESTAND 3 TIER (£174,485), PAPER CRAFT LITTLE BIRDIE (£168,470), and WHITE HANGING HEART T-LIGHT HOLDER (£106,293) lead.
- **Notable overlap:** REGENCY CAKESTAND 3 TIER is simultaneously a top-3 revenue product and the most-returned real product (181 returns), a fragile, high-value item is a natural candidate for packaging or quality investigation rather than customer dissatisfaction.

- **Category tagging limitation:** The keyword-based category tagging (Set / Bag / Others) placed 86.7% of all sales in “Others,” which confirms this approach is too coarse to drive real category strategy.

5.3 Customer Analytics

- **Fixed-threshold segmentation breaks down:** The fixed-threshold segmentation (customers above £100,000 = High, above £50,000 = Medium) classified 4,354 of 4,373 customers (99.6%) as “Low.” This isn't a finding about the customers — it's a sign the fixed cutoffs don't fit this revenue distribution, where a small number of very large accounts skew any absolute threshold.
- **Percentile segmentation is far more informative:** A percentile-based approach tells a much more usable story: the top 5% of customers (218 people, excluding unidentified guest transactions) generated £4.03M — 37.8% of total revenue.
- **Extreme customer concentration:** The single highest-spending customer (ID 14646) generated £279,489 — over 125× the average customer's spend.
- **One-time buyers:** 30.0% of customers made exactly one purchase and never returned; 69.97% of customers made a repeat purchase, these two figures are internally consistent ($100\% - 30.0\% \approx 69.97\%$), which is a useful sanity check on the underlying query logic.
- **Apparent churn (caveat):** 55.3% of customers had no recorded purchase after late-October 2011. This should be read cautiously: the dataset itself ends in December 2011, so a portion of this figure reflects the observation window closing rather than confirmed churn.

5.4 Returns & Cancellations

- **Cancellation rate:** Nearly 1 in 5 invoices (19.97%) were cancelled, costing an estimated £896,812 , roughly 8.4% of gross revenue.
- **Largest-order risk:** The single largest order in the dataset (£168,470) was cancelled in full immediately after being placed, a reminder that a small number of large-order cancellations can materially move the total loss figure.
- **Return-prone products:** After excluding admin/adjustment entries, real returned products are dominated by fragile or multi-piece items, REGENCY CAKESTAND 3 TIER, JAM MAKING SET WITH JARS, SET OF 3 CAKE TINS, consistent with shipping-damage risk rather than a broad quality problem.
- **Timing of returns:** Returns track overall order volume closely (highest on Thursday, concentrated 10am–4pm, 82% from the UK) , there's no distinct “high-risk day/time,” the pattern simply mirrors when orders happen.

5.5 Geographic Trends

- **UK dominance:** The United Kingdom accounts for £8.19M (76.9%) of revenue, 4.26M units, and 3,951 of 4,373 customers (90.4%) , this is fundamentally a UK-concentrated business with a modest international tail.
- **International AOV:** The Netherlands has the highest average order value (£120.06) despite comparatively low volume, consistent with a small number of high-value international (likely business) accounts rather than broad consumer demand.
- **Low international return share:** International returns are proportionally minimal (Germany 453, EIRE 247, France 149, versus 7,533 in the UK).

5.6 Order Patterns

- The average invoice contains roughly 21 distinct products but nearly 200 total units, a gap that indicates multi-unit, bulk-style purchasing rather than typical single-consumer basket behaviour, again consistent with the wholesale signal.

6. Business Recommendations

1. Audit the top cancelled invoices and the customers behind them to determine whether the 19.97% cancellation rate is driven by stock-outs, pricing errors, or large wholesale orders being backed out, recovering even a fraction of the £897K at risk is a concrete, quantifiable win.
2. Replace fixed-value customer tiers with percentile-based segmentation, and build a dedicated retention/account-management track for the top 5% (218 customers, 37.8% of revenue), losing even a few of these accounts would materially move total revenue.
3. Investigate REGENCY CAKESTAND 3 TIER and other fragile, high-return, high-revenue products for packaging or handling fixes before assuming a demand or quality problem.
4. Launch a second-purchase incentive aimed at first-time buyers within 60–90 days of their first order, targeting the 30% one-time-buyer segment before they lapse.
5. Fix the upstream data pipeline so admin/adjustment line items (Manual, POSTAGE, Discount, etc.) are flagged at source rather than mixed into product-level reporting, and so customer ID capture is more complete (reducing the custid = 0 pool).

7. Advanced Analytics Performed

- RFM Analysis (Recency, Frequency, Monetary) at the customer level
- Percentile-based VIP identification (top 5% via NTILE)
- Repeat purchase cycle / purchase-gap detection using window functions (LAG)
- Cohort identification by first purchase month
- Weekday/hourly time-series decomposition of sales and returns
- Country-level performance and return-rate benchmarking

8. Limitations

- The dataset spans roughly 13 months (Dec 2010–Dec 2011), covering a single holiday cycle — seasonality and churn conclusions should be validated against a longer time horizon before being treated as stable patterns.
- Product categorisation is keyword-based (Set / Bag / Others) rather than a true taxonomy, and placing the large majority of products in “Others”, category-level strategy would need better source data.
- “Churn” as defined here (no purchase since late Oct 2011) is confounded with the dataset's own end date; a portion of the 55.3% figure likely reflects the observation window closing, not confirmed customer loss.

9. Future Scope

The current analysis is descriptive and explains what happened. The planned next phase adds a predictive layer:

- A churn prediction model (Random Forest / Logistic Regression baseline) using RFM-derived features, evaluated on precision, recall, and AUC-ROC.

- K-means clustering on RFM features to validate (or challenge) the manual customer segments in Section 5.3.
- Market basket analysis to properly quantify product affinity, replacing the earlier SQL self-join approach for both correctness and performance.
- Lightweight time-series forecasting of monthly revenue.

10. Tech Stack

- MySQL 8 — data modeling, querying, window functions
- Python (Pandas) — data cleaning, EDA
- Excel — consolidated query output for cross-checking and inference

11. What I Learned

- *Handling messy, real-world transactional data*: inconsistent formatting, adjustment entries, and missing identifiers and treating that mess as a finding in itself rather than something to silently clean away.
- *Writing analytical SQL beyond basic SELECT statements*: window functions, CTEs, and percentile-based ranking for genuinely behavioural questions.
- *The difference between a query that runs and a query that's right*, several of the queries above only became trustworthy after cross-checking outputs against each other (e.g. one-time-buyer rate against repeat-purchase rate) and against domain logic (e.g. noticing “Manual” topping a product ranking).
- *Translating raw metrics into a business narrative* : the wholesale-behaviour pattern only became visible by reading several metrics together, not from any single query.